BENG 203 Assignment 3

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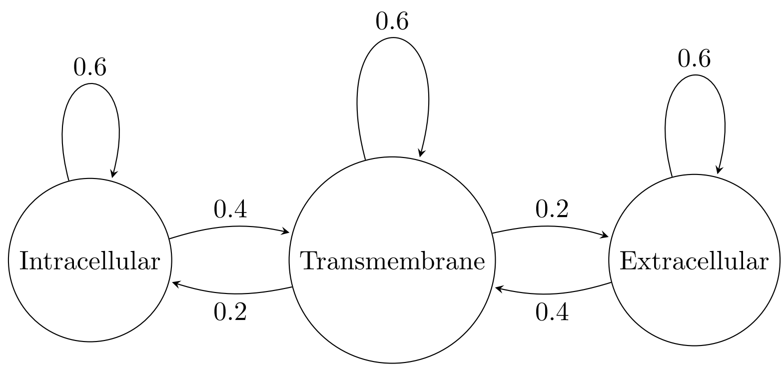
Problem 1:



For this Hidden Markov Model, there are three hidden states $Q = \{Intracellular, Transmembrane, Extracellular\}$. The observed states Σ are the 20 amino acids. The initial state distribution is $\pi = \{1, 0, 0\}$ since we start in the intracellular state. The emission probabilities vary by state:

$$e_I(a) = \frac{1}{20} \qquad \forall a \in \Sigma$$

$$e_T(a) = \begin{cases} \frac{15}{100}, & a \in \{L, I, V, F, W\} \\ \frac{1}{100}, & \text{otherwise} \end{cases}$$

$$e_E(a) = \begin{cases} \frac{43}{300}, & a \in \{R, K, D, E, Q, N\} \\ \frac{1}{100}, & \text{otherwise} \end{cases}$$

The transition probabilities are given by the edges in the diagram, which can be represented in matrix form as follows:

$$T = \begin{bmatrix} 0.6 & 0.4 & 0 \\ 0.2 & 0.6 & 0.2 \\ 0 & 0.4 & 0.6 \end{bmatrix}$$

Problem 2:

(a) The theoretical residue mass is: 781.349 Da.
The theoretical peptide mass is: 799.365 Da.
The ion mass for charge $Z = 2$ is: 400.690 Da.

(b) $PEPMASS = 777.25439453 = \frac{m}{z}$ and $CHARGE = 7+ = z$ are given in the mgf file.

$$M_{peptide} = z\left(\frac{m}{z}\right) - zM_{proton} = 7(777.25439453) - 7(1.007276466812) = 5433.7298$$

$$M_{residue} = M_{peptide} - M_{water} = 5433.7298 - 18.0153 = 5415.7145$$

High mass accuracy is important because it narrows the number of possible peptide candidates, making peptide identification is more specific and reducing false matches. It also improves confidence in matching experimental precursor and fragment ions to their theoretical masses.

(c)

$$\text{ppm error} = \frac{|m_{observed} - m_{theoretical}|}{m_{theoretical}} \times 10^6$$

Only one candidate peptide was found in the given tolerance:

QQPDGLAVVGVFLKVGDANPALQKVLDALDSIKTKGKSTDFPNFDPGSLLPN

This peptide occurs at positions 135–186, has theoretical neutral mass 5433.8871 Da, and has ppm error 28.95 ppm. The second closest candidate peptide is:

LKEPISVSSQQMLKFRTLNFNAEGEPELLMLANWRPAQPLKNRQVRG

This peptide is slightly out of tolerance at 30.68 ppm, and occurs at positions 211–257 with theoretical neutral mass 5433.8966 Da. I have included it here for completeness as it may be due rounding errors from the amino acid masses I am using.

(d) After generating the theoretical spectrum, the charge 2 ions for each candidate were output to a csv. Due to the size of the table, it has been left as a separate csv rather than embedded in this document. See candidate_1_spectrum.csv and candidate_2_spectrum.csv for the results, which have been validated against Protein Prospector's theoretical spectrum generator.

(e) To make a spectral similarity score between the experimental spectrum and the theoretical spectrum, I would use a cosine similarity score. First, I would match each theoretical fragment ion to the closest experimental peak with a small fragment mass tolerance, such as 10 ppm. This tight tolerance is chosen as the MS2 fragments were measured with very high mass accuracy, so we can be confident in the matches. Then, I would normalize the intensities of the experimental peaks and assign the theoretical peaks all to an intensity of 1 since we don't have true abundance measurements. Now, the spectra can be represented as vectors, and the cosine similarity can be calculated as follows:

$$\text{Cosine Similarity} = \frac{A \cdot B}{\|A\|\|B\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}}$$

Where A is the vector of theoretical peak intensities and B is the vector of matched experimental peak intensities.
Inputs: theoretical spectrum (m/z, ion label), experimental spectrum (m/z, intensity), fragment mass tolerance.
Output: A score between 0 and 1, where 1 indicates a strong match between the theoretical and experimental spectra, and 0 indicates no similarity.

Algorithm 1 Cosine Similarity Scoring

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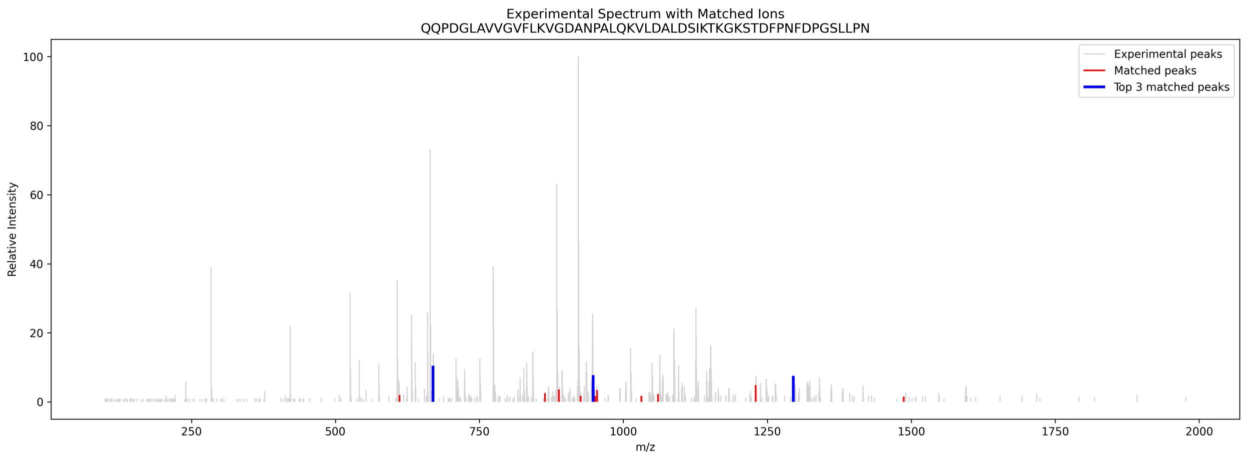
Inputs:
    theoretical_spectrum = list of (m/z, ion label)
    experimental_spectrum = list of (m/z, intensity)
    fragment_mass_tolerance = 10 ppm
Output:
    cosine_similarity_score
max_intensity ← max(experimental intensities)
for each peak e in experimental_spectrum do
    e.intensity ← e.intensity/max_intensity
end for
A ← [ ]
B ← [ ]
for each theoretical peak t in theoretical_spectrum do
    match ← closest experimental peak within 10 ppm of t
    append(A, 1)
    if match exists then
        append(B, match.intensity)
    else
        append(B, 0)
    end if
end for
numerator ← A · B
denominator ← ‖A‖ × ‖B‖
cosine_similarity_score ← numerator / denominator
return cosine_similarity_score
```

(f) Peptide: QQPDGLAVVGVFLKVGDANPALQKVLDALDSIKTKGKSTDFPNFDPGSLLPN
Top 3 most intense matched peaks:

(i) Experimental m/z = 669.6343
Ion = c26
Charge = 4
Sequence = QQPDGLAVVGVFLKVGDANPALQKVL
PPM error = 3.37

(ii) Experimental m/z = 947.5417
Ion = a37
Charge = 4
Sequence = QQPDGLAVVGVFLKVGDANPALQKVLDALDSIKTKGK
PPM error = 0.82

(iii) Experimental m/z = 1295.2152
Ion = a50
Charge = 4
Sequence = QQPDGLAVVGVFLKVGDANPALQKVLDALDSIKTKGKSTDFPNFDPGSLL
PPM error = 9.75


Experimental Spectrum with Matched Ions
QQPDGLAVVGVFLKVGDANPALQKVLDALDSIKTKGKSTDFPNFDPGSLLPN
Experimental peaks
Matched peaks
Top 3 matched peaks
Relative Intensity
m/z
0
20
40
60
80
100
250
500
750
1000
1250
1500
1750
2000